



## Department of Electrical and Electronics Engineering Academic Year 2021 – 2022 (Even Semester)

**Degree, Semester & Branch:** IV Semester B.E. EEE B

**Course Code & Title:** EE8402 Transmission and Distribution

**Name of the Faculty member :** Dr.B.Deepa Lakshmi

### Innovative Practice Description

- **Unit / Topic:** Unit I / Types of Transmission line and Distribution line conductors
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** One Minute Paper

- **Justification:**

The chosen topic - types of conductors have 3 types in transmission and distribution conductors. Since each type has its own definition and limitation. After teaching the concept, I thought of conducting this activity for enabling the students to give the difference between each type of conductors which enhance their learning level and as a teacher I can judge the understanding level of the students.

**Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

After completing the topic, I gave 5 minutes for the students to think about the topic without writing anything.

Total Strength: 30

Reporter : Myself

At the end of the class

- ✓ I asked the students to think about various types of conductors in transmission and distribution lines for 5 minutes.
- ✓ I told them to write as much as they remember for 3 minutes.
- ✓ Finally, I collected the papers from the students.(2 minutes)

- **CO – PO / PSO mapping:**

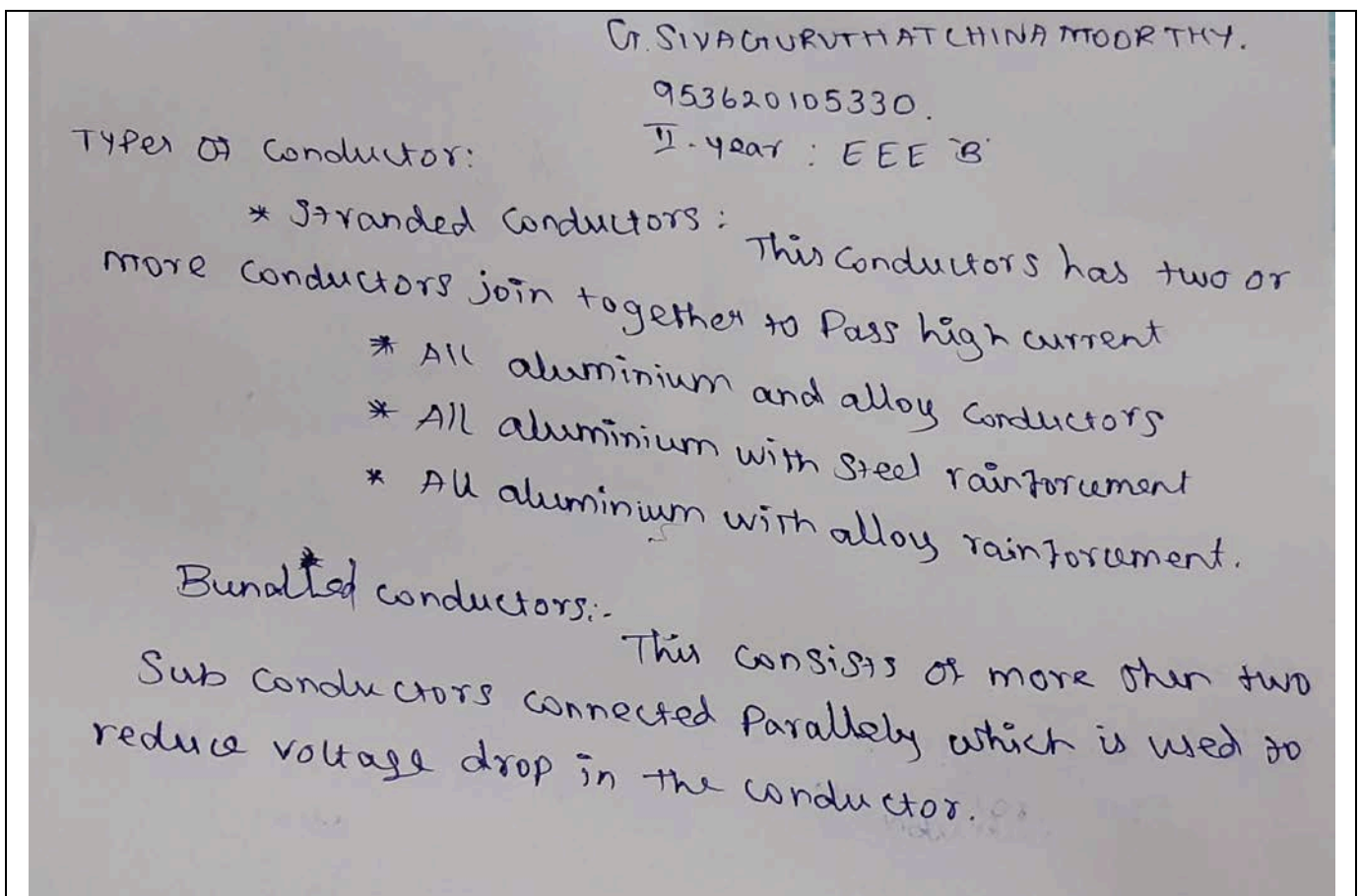
| CO  | PO1 | PO2 | PO3 | PO8 | PO9 | PO10 | PO12 | PSO1 |
|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 3   | 2   | 2   | 1   | 1   | 1    | 2    | 2    |



- PO / PSO mapped:

| Innovative practice           | PO10                                                                                                  |
|-------------------------------|-------------------------------------------------------------------------------------------------------|
|                               | 1                                                                                                     |
| Justification for correlation | Due to this activity, student's written communication will be enhanced. So it is slightly correlated. |

- Screenshot of the activity:





Types of Stranded Conductors:

K. Rajkumar

953620105321

II Year - EEE 'B'

AAC - All Aluminium Conductor

AAAC - All Aluminium with alloy conductor

ACSR - Aluminium Conductor with Steel reinforcement

ACAR = Aluminium Conductor with Alloy reinforcement.

## Reflective Critique:

### ❖ *Feedback of practice from students and other stakeholders:*

Students felt that writing about the given topic make them understand it well and able to recollect immediately.

### ❖ *Benefit of the practice:*

1. All the students were able to write the given topic.
2. After conducting the activity, I came to know that students understood the topic and they were able to explain it.

### ❖ *Challenges faced in implementation:*

1. Time utilization for conducting the activity
2. Some students could not be able to recollect and write immediately.

## References:

- ✓ D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Second Edition, 2008.
- ✓ V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013
- ✓ <https://oncourseworkshop.com/self-awareness/one-minute-paper/>

Signature of Faculty Member

HOD



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### Innovative Practice Description

- **Unit / Topic:** Unit II / Corona
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen:** Think Pair Share

- **Justification:**

The chosen topic – corona provides safety to the transmission lines and reduces flash over. After teaching the concept, I thought of conducting this activity for making the students to provide knowledge about corona and its effects which enhance the learning level and as a teacher I can judge the understanding level of the students.

**Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

After completing the topic, the students will be paired with their neighbours, 4 students as a group.

Total Strength is 30, Number of Pairs – 8

Photographer: Myself

Reporter: Myself

At the end the Class (Last 10 minutes)

- I asked the students to **think** about corona, factors affecting corona and its loss for 2 minutes.
- Then I told them to **Pair** with their neighbours and discuss about the concept of corona in transmission line for another 2 minutes.
- Finally, I selected 1 Pair from each column randomly and ask them to **share** about corona and its advantages. (6 minutes)
- Finally, I summarized the points again about corona in transmission line.



- CO – PO / PSO mapping:

| CO  | PO1 | PO2 | PO3 | PO8 | PO9 | PO10 | PO12 | PSO1 |
|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 3   | 3   | 3   | 1   | 1   | 1    | 2    | 2    |

- PO / PSO mapped:

| Innovative practice           | PO9                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------|
|                               | 1                                                                                            |
| Justification for correlation | Due to this activity, student's team work is getting enhanced. So it is slightly correlated. |

- Screenshot of the activity:







## Reflective Critique:

### ❖ *Feedback of practice from students and other stakeholders:*

Students felt that discussing with the team members and sharing the concept with everyone make them understand it well and enhanced their knowledge.

### ❖ *Benefit of the practice:*

1. All the students were able to share the topic given in an effective way.
2. Students understood the concept which was reflected from their answers during sharing session.

### ❖ *Challenges faced in implementation:*

1. Time utilization for conducting the activity
2. Slow learners were not able to recollect and present some points during sharing session.

## References:

- ✓ D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Second Edition, 2008.



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- ✓ V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013
- ✓ <https://www.readingrockets.org/strategies/think-pair-share>

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### Innovative Practice Description

- **Unit / Topic:** Unit III / Types of Towers
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO6
- **Activity Chosen:** Visible Quiz

- **Justification:**

The chosen topic - types of towers have 4 types of line supports for overhead line conductors. Since each type has its own definition and limitation, after teaching the concept, I thought of conducting this activity for enabling the students to differentiate between each types of line supports which enhance their learning level and as a teacher I can judge the understanding level of the students.

**Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

After completing the topic, I gave 3 minutes for the students to think about the topic without writing anything.

Total Strength: 30

Photographer: Myself

At the end of the class

- ✓ I asked the students to think about various types of towers for 3 minutes.
- ✓ I asked them questions with multiple choice (A,B,C,D) by providing set of large cards imprinted with one of the four letters 5 minutes.
- ✓ Finally, I summarized the questions and answers to the students and cleared their doubts.(2 minutes)

- **CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO3 | PO4 | PO8 | PO9 | PO10 | PO12 | PSO1 |
|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO3 | 3   | 3   | 3   | 1   | 1   | 1   | 1    | 2    | 2    |





- PO / PSO mapped:

| Innovative practice           | PO7                                                                                                                                           |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|                               | 1                                                                                                                                             |
| Justification for correlation | Due to this activity, students will understand the impact of professional engineering solution for environment. So it is slightly correlated. |

- Screenshot of the activity:





## Reflective Critique:

### ❖ *Feedback of practice from students and other stakeholders:*

Students felt that answering the options about the topic by using cards is understandable & interesting and able to recollect it.

### ❖ *Benefit of the practice:*

1. All the students were able to answer the options given for the question.
2. After conducting the activity, I came to know that students understood the topic and they were able to explain it.

### ❖ *Challenges faced in implementation:*

1. Time utilization for conducting the activity
2. Some students could not be able to recollect and answer immediately.

## References:

- ✓ D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Second Edition, 2008.
- ✓ V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013
- ✓ [https://www.ideaedu.org/Portals/0/Uploads/Documents/IDEA%20Papers/IDEA%20Papers/PaperIDEA\\_53.pdf](https://www.ideaedu.org/Portals/0/Uploads/Documents/IDEA%20Papers/IDEA%20Papers/PaperIDEA_53.pdf)

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**Innovative Practice Description**

- **Unit / Topic:** Unit V / Substation Layout and Methods of Grounding
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO15
- **Activity Chosen:** Field Visit
- **Justification:**

The topic Substation Layout and Methods of Grounding has different types and used for various applications. After completing the topic, I thought of taking the students for field visit to enable the students to acquire more knowledge in observing the layout and grounding techniques.

**Time Allotted for the Activity:** 45 Minutes

- **Details of the Implementation:**

After completing the topic, I took the students to the RIT – Power House and shown them how the grounding is done and explained about the input power and power distribution inside the college.

Total Strength: 30

Photographer: one student - Mr. Suryaprakash

At the end of the visit

- ✓ I asked some students to measure the earth resistance. Some students voluntarily measured the resistance.
- ✓ I request all the students to discuss among themselves for 5 minutes.
- ✓ Finally, I asked questions to some students about the substation layout and methods of grounding and the students answered well which shows the understanding of the topic through this field visit.(5 minutes)



- CO – PO / PSO mapping:

| CO  | PO1 | PO2 | PO3 | PO6 | PO8 | PO9 | PO10 | PO12 | PSO1 |
|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 3   | 3   | 3   | 2   | 1   | 1   | 1    | 2    | 2    |

- PO / PSO mapped:

| Innovative practice           | PO7                                                                                                                                           |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|                               | 1                                                                                                                                             |
| Justification for correlation | Due to this activity, students will understand the impact of professional engineering solution for environment. So it is slightly correlated. |

- Screenshot of the activity:





## Reflective Critique:

### ❖ *Feedback of practice from students and other stakeholders:*

Students were interested in attending this type of field visit and requested to conduct more such activities in future.

### ❖ *Benefit of the practice:*

- ✓ The assessment of effectiveness of the activity was good in a way that while asking questions, most of the students answered and shared many points.
- ✓ After conducting the activity, I understood that the students could be able to recollect the points about substation and grounding methods.

### ❖ *Challenges faced in implementation:*

- ✓ Time is not enough for this activity, so will allot more time in future.

## References:

- ❖ D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Second Edition, 2008.
- ❖ V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013

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